Report of PhD activities in the academic year 2017 / 2018

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October 7, 2018

Abstract

This document describes the achievements, current state and future plans of the research project, moreover it contains the courses and conferences attended in the 2017 / 2018 academic year.

Contents

1 Current state of the research topic - Atmosphere modelling

1.1 The shallow domain concept

Considered as a geophysical fluid, the polluted atmosphere shares the shallow domain characteristics with other natural large-scale fluids such as seas and oceans. This means that its domain is excessively greater horizontally than in the vertical dimension, leading to the classic hydrostatic approximation of the Navier-Stokes equations. The authors of the [Azérad and Guillén, 2001] article have proved a convergence theorem for this model with respect to the ocean, without considering pollution effects. The novelty achieved this year in this project is to provide a generalisation of their result translated to the atmosphere, extending the fluid velocity equations with an additional convection-diffusion equation representing pollutants in the atmosphere.

The model we constructed utilises classical concepts as its foundation and it merges them with relatively new or more specific ideas. On the one hand, we use general and well-known approaches such as the wind flow being modelled by the incompressible anisotropic Navier-Stokes equations, and using a convection-diffusion equation to describe the pollution effect. On the other hand, the model also incorporates notions like taking advantage of a downwind-direction matching coordinate system, using a Gaussian pulse as a source term, and rescaling the diffusivity parameters according to the largely different horizontal and vertical scales.

The equations governing the air velocity $\mathbf{v}$ and pressure q in the atmosphere are the incompressible Navier-Stokes equations, in which we use different viscosities according to vertical or horizontal directions, and the density is taken to be identically equal to one:

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} - \Delta_\nu \mathbf{v} + \nabla q + 2\mathbf{w} \times \mathbf{v} = \mathbf{g} \text{ in } \Omega_\epsilon \times (0,T) \tag{1}$$

$$\nabla \cdot \mathbf{v} = 0 \text{ in } \Omega_\epsilon \times (0,T) \tag{2}$$

Contaminants emitted into the atmosphere are transported by the wind, mixed and dispersed by turbulent behavior, and deposited onto the ground by gravity. The dispersion of pollutants is commonly modeled by a diffusive equation of the form

$$\partial_t C + \mathbf{v} \cdot \nabla C = K_x \partial^2_{xx} C + K_y \partial^2_{yy} C + K_z \partial^2_{zz} C + S, \tag{3}$$

where C is the pollution concentration.

One of the main tools that we use in order to exploit the "almost" two-dimensional structure of the atmosphere is to introduce the aspect ratio

$$\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}},$$

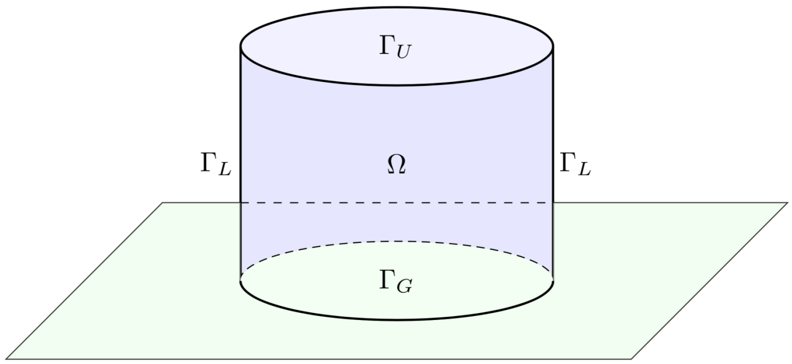

Figure 1: A local slice of the atmosphere

which incorporates the shallowness in a mathematical way. This makes a natural scaling process possible: we use the $x = x_1, y = x_2, z = \epsilon x_3$ coordinates on a new, ϵ-independent domain. Correspondingly, we rescale the kinematic terms, the viscosity terms and the diffusivity coefficients as well.

$$v_1 = u_1^\epsilon, \quad v_2 = u_2^\epsilon, \quad v_3 = \epsilon u_3^\epsilon, \quad p = p^\epsilon,$$
$$\nu_x = \nu_1, \quad \nu_y = \nu_2, \quad \nu_z = \epsilon^2 \nu_3,$$
$$K_y = K_2, \quad K_z = \epsilon^2 K_3.$$

As we have referred to it earlier, we choose to fix the coordinate system in a way so that x and y match the downwind and crosswind direction, respectively (while z represents the vertical dimension). Thus, as a second step of the rescaling process, we can also exploit the fact that we chose the coordinate system in a way so that the x axis matches the downwind direction. In this case we use an additional scaling equation, which is different in nature from the previous ones in the sense that it is not derived from the shallowness of the domain but rather suggested by the fact that in the downwind direction it is natural to assume that the diffusion is negligible compared to downwind transport, i.e.

$$|u_1 \partial_{x_1} C| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right|,$$

which leads us to using

$$K_x = \epsilon K_1.$$

We use a local cartesian coordinate system so that we can neglect the curvature of the Earth.

Here we mention that the challenge of the local area model is to define open boundary conditions on the lateral boundary. We use a fix zero approach on the upper and lateral boundaries – the concentration being set to zero actually stands for a constant concentration with the background flow being removed.

Finally, for modelling the source function of the concentration we use the Gaussian impulse approach. We use a parametrised nascent delta function to describe the source term, with the parameter being ϵ. We have $S_\epsilon(t, t_s, x, x_s) = s(t, t_s)\delta_\epsilon(x - x_s)$, with the activation term $s(t, t_s) = IH(t - t_s)$ where $H(\cdot)$ is the Heaviside step function. In the limit model we naturally arrive to the delta distribution as a source.

1.2 The anisotropic Navier-Stokes equations and the primitive equations

Putting together all the concepts and ideas described in the previous section, we arrive to two sets of equations that we will need in order to present the main theorem. The first set of equations (anisotropic case) uses the Navier-Stokes equations in all three dimensions.

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon\beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\epsilon^2\{\partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon\} - \epsilon\beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0, T)$$

$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0, T)$$

$$\mathbf{u}^\epsilon(\cdot, t = 0) = \mathbf{u}_0, C^\epsilon(\cdot, t = 0) = C_0^\epsilon \text{ in } \Omega$$

$$C^\epsilon = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C^\epsilon = \partial_3 C^\epsilon = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C^\epsilon \in L^2 L^2(0, T; \Gamma_G)$$

The hydrostatic model, on the other hand, is obtained by formally taking the limit as ϵ vanishes both in the velocity and pollution equations. The limit model is made of the following equations.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0,T)$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0,T)$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0,T)$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0,T)$$

$$u_i(\cdot, t=0) = u_{0i}, C(\cdot, t=0) = C_0 \text{ in } \Omega, i = 1{,}2$$

$$C = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0,T), \partial_2 C = \partial_3 C = 0 \text{ on } \Gamma_G \times (0,T), \partial_1 C \in L^2 L^2(0,T;\Gamma_G)$$

1.3 Main result

The main theorem describes a more meaningful connection between these two sets of equations. It says that not only is it true that we formally get the hydrostatic model from the original one by taking the limit in the equations, but also, we show that the weak solutions of the anisotropic model converge to a weak solution of the hydrostatic limit model — in other words, taking the limit we arrive to a justification of the hydrostatic model for the case of the polluted atmosphere, where, instead of the originally used wind momentum equation, the hydrostatic air pressure assumption is used to describe the process in the vertical dimension.

Theorem 1. *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$; then as the aspect ratio ϵ tends to zero, any weak solution $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic equations converge to a weak solution ($\mathbf{u}$, C) of the hydrostatic equations of the polluted atmosphere.*

The beginning step of the proof is to obtain the following energy inequality that yields priori estimates such as $L_t^\infty L_x^2$ boundedness for C^ϵ; $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$.

$$
\begin{aligned}
&\frac{1}{2}\big(\big\|\mathbf{u}_H^\epsilon(t)\big\|_{L^2}^2 + \epsilon^2\big\|u_3^\epsilon(t)\big\|_{L^2}^2 + \big\|C^\epsilon(t)\big\|_{L^2}^2\big) + \int_0^t \big[\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\big]\,\mathrm{d}\tau \\
&+ \int_0^t \Big[(\epsilon K_1 - \zeta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_1 C^\epsilon\|_{L^2}^2 + (K_2 - \zeta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_2 C^\epsilon\|_{L^2}^2 \\
&+ (K_3 - \zeta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_3 C^\epsilon\|_{L^2}^2\Big]\,\mathrm{d}\tau \\
&\le \frac{1}{\zeta}K_1\frac{\epsilon}{2}\int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 + \frac{1}{2}\big(\big\|\mathbf{u}_H^\epsilon(0)\big\|_{L^2}^2 + \epsilon^2\big\|u_3^\epsilon(0)\big\|_{L^2}^2 + \big\|C^\epsilon(0)\big\|_{L^2}^2\big) \\
&+ \int_0^t (S^\epsilon, C^\epsilon)\,\mathrm{d}\tau.
\end{aligned}
\tag{4}
$$

Once we have boundedness, we have the weak convergence of the linear terms, while the weak convergence of the nonlinear terms not including the pollution term had been shown before by [Azérad and Guillén, 2001]. We prove the convergence of the term $(\mathbf{u}^\epsilon C^\epsilon, \nabla\tilde{C})$ by using a compactness result. We show that $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$ following these two steps:

- we have a bound for the time derivative of the concentration:

$$
\frac{\partial C^\epsilon}{\partial t} = -u^\epsilon \nabla C^\epsilon + \sqrt{\epsilon}K_1\partial_1\partial_1\sqrt{\epsilon}C^\epsilon + K_2\partial_2\partial_2 C^\epsilon + K_3\partial_3\partial_3 C^\epsilon - S^\epsilon,
$$

  Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$,

- we have an equicontinuity for velocity: $\big\|u_i^n - u_i^n(\cdot + \xi, t)\big\|_{L^2L^2} \to 0$ as $\xi \to 0$.

Combining these results allow us to apply a compactness theorem that yields the convergence.

Additionally, we also provide an existence theorem for the limit model.

1.4 Outlook

We plan to use a diffusion matrix instead of the original coefficients, investigate the case of strong convergence, and visualise the results by the finite element method.

2 Courses, summer schools and seminars

2.1 Courses

2.1.1 Physics of the atmosphere - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila

This course gave an overview of selected topics related to the dynamics of the atmosphere. The general characteristics of the atmosphere were described in terms of mass and chemical composition. Gases with the highest mixing ratio are nitrogen, oxygen and argon - it is pointed out that these are constants, and the variabilities are provided by the remainder gases which are present in much smaller percentage: trace gases and aerosols. Although their lifetime is small, these later's effect is huge as they are able to capture radiation and alter energy balance. A big chapter in this course is represented by the thermodynamics of the atmosphere and gas laws. The ideal gas equation is considered in different forms, as well as the concept of hydrostatic balance and geopotential. The next step is the first law of thermodynamics in different approaches, and the notion of an adiabatic process. The main ideas and building blocks of chemical and photochemical reactions (thermal decomposition, photolysis) in the atmosphere are introduced as well as the notion of the rate of reaction, and several chemical reaction equations are solved in this chapter.

2.1.2 Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI

This course gave an overview on shock capturing schemes for the numerical solution of systems of conservation and balance laws. The emphasis was on the construction of high order finite volume and finite difference schemes, which are based on the three main ingredients: numerical flux, high order non oscillatory reconstruction, and high order integration in time. The first part of the course illustrated some models in which the nonlinearity produces shocks that propagate in time. Simple three point schemes, such as upwind, Lax-Friedrichs, and Lax-Wendroff was reviewed, and their behaviour on linear equations and systems analysed, before the illustration of more complex methods for the quasilinear systems. The step by step construction of the schemes was illustrated in lab sessions, and the methods were applied to concrete examples of physical interest. The last part of the course was devoted to systems with source term.

2.2 Conferences, seminars, summer schools

2.2.1 Intensive Program on Fluids and Waves, GSSI

This training activity within the European Innovating Training Network ModCompShock surveyed recent results in the analysis of fluids and waves with emphasis on modelling and computational methods as well as its theoretical aspects.

a) Week 1. Workshop/School in Numerics for Fluids and Waves

b) Week 2. Mini courses by J. Bedrossian. and D. Cordoba

c) Week 3. Mini courses by C. Bardos, Z. Grujic, J. Malek

d) Week 4. Workshop on Mathematical fine structures in fluid dynamics

2.2.2 Women in applied and computational Mathematics, GSSI

This three-day event was designed for researchers challenged by the application of various techniques in modern analysis and computational mathematics to intriguing problems arising in physics, material science and biology. The topic and the results of this present research project were represented - together with the work of many early-stage female researchers - in the poster session.

2.2.3 Evans function methods in the stability analysis of periodic wavetrains, Prof. R. Plaza

This lecture series provided an introduction to recent techniques in the spectral stability analysis of spatially periodic travelling waves which involve the periodic Evans function introduced by Gardner.

2.2.4 On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza

2.2.5 Secondment at the University of Sussex, Prof. Omar Lakkis

The aim of this 6-week secondment was to create both the theoretical and technical foundations for numerical simulations in order to visualise the results of the project. The main results and steps are the following:

a) Software packages such as Docker, Anaconda, ParaView, FEniCS and Alberta have been installed

b) Writing scripts to login / connect to the HPC server at the Sussex University

c) Getting to know FEniCS: tutorial files and the FEniCS version of the "hello world" programs

d) Creating a basis for future work with specific theoretical details that arise when implementing this project - such as the loss of incompressibility when it comes to the FEM approach, and the use of Lagrange multipliers.

2.2.6 Waves in Flows Summer School in Prague

The aim of this summer school was to present a comprehensive series of lectures on various aspects of waves and their role in mathematical analysis and numerical simulation of different types of flows. The school was organised as a multidisciplinary event with topics ranging from theoretical and numerical mathematics up to physics and applications. The emphasise was on models related to topics originating in environmental, biomedical and industrial applications. Besides of the course lectures, the program included a one day workshop dedicated to short lectures given by attending scientists.

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